# MUHAMMAD DANISH

AI Engineer | Python Backend | Agentic Systems

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## PROFESSIONAL SUMMARY

AI Engineer and Full-Stack Developer with hands-on experience building agentic AI systems, multi-agent automation platforms, LLM-driven workflows, and real-time backend microservices. Proficient in Python, FastAPI, Django, LangChain-style orchestration, and RAG architectures. Built production-grade AI tools including a multi-agent coordination platform (Council AI) and a CNN-based computer vision model. Strong background in Linux server deployment, REST API development, and integrating third-party AI models. Available to start immediately and comfortable working on-site in Lahore.

## EXPERIENCE

### VoIP Support Engineer & AI Automation Developer

Abacus Consulting, Lahore | *Jul 2025 – Present*

- Developed Council AI — a multi-agent automation platform coordinating specialized AI bots for research, data analysis, and workflow automation across business processes
- Built Line QA Bot — an AI assistant using LLM-driven analysis of call logs and quality metrics to auto-detect failure patterns and generate structured QA reports
- Developed backend microservices in Python (FastAPI, Django) to support high-concurrency requests in live production environments
- Integrated REST APIs and third-party services into unified software ecosystems; experienced with OpenAI API integration
- Deployed and managed applications on Linux servers (Ubuntu, AlmaLinux) using SSH, Nginx, Docker, and Bash automation scripts
- Automated operational workflows using Python and Bash scripting, reducing manual effort by 40%
- Administered real-time infrastructure for 150+ concurrent users; maintained 99.9% uptime under SLA-driven conditions

## KEY AI PROJECTS

**Council AI — Multi-Agent Automation Platform**

- Designed and built a multi-agent AI system coordinating specialized autonomous bots for research, data analysis, and business workflow automation
- Implemented agent orchestration logic with task delegation, inter-agent communication, and result aggregation — directly analogous to LangChain/CrewAI agentic patterns
- Built in Python with FastAPI backend; designed for extensibility to plug in different LLM providers

**Industrial Defect Detection — Computer Vision Model (GDXray)**

- Developed a CNN-based binary classifier in Python (PyTorch/TensorFlow) for detecting casting and weld defects in X-ray images
- Applied data augmentation (rotation, flip, brightness) to improve generalization on limited samples; optimized using accuracy, recall, and F1-score
- Demonstrates hands-on experience with model training, fine-tuning, and evaluation — applicable to edge-computing and spatial AI use cases

**IoT Security Monitoring Hub — Real-Time Backend System**

- Built full-stack real-time IoT dashboard using Django REST Framework, React, MQTT (Mosquitto), Redis, and WebSockets — processing 100+ messages/min at <1s latency
- Deployed on Ubuntu with Nginx and ASGI (Daphne); implemented JWT auth and REST APIs — architecture directly transferable to high-concurrency AI microservices

**Line QA Bot — LLM-Driven Analysis Tool**

- AI assistant that ingests structured call log data, applies LLM-based pattern detection, and auto-generates QA reports — implements core RAG-style data retrieval and generation pipeline
- Built with Python backend; designed to interface with LLM APIs for inference

## TECHNICAL SKILLS

**AI & ML:** Python (PyTorch, TensorFlow), CNNs, model fine-tuning, data augmentation, prompt engineering, LLM API integration (OpenAI, Anthropic, Gemini)
**Agentic AI:** Multi-agent orchestration, agentic workflow design, RAG architecture, task delegation pipelines — hands-on via Council AI project
**Backend:** FastAPI, Django, Flask, REST APIs, WebSockets, Redis, MySQL, SQL, high-concurrency microservices
**Languages:** Python, JavaScript, C/C++, Bash, SQL
**DevOps & Cloud:** Linux (Ubuntu, AlmaLinux, openSUSE), Docker, Nginx, SSH, Git, cloud VPS deployment (Contabo/AWS-compatible)
**Bonus Skills:** Computer Vision (OpenCV, CNN-based detection), real-time data pipelines (MQTT, WebSockets), Voice/Audio processing familiarity via VoIP work

## EDUCATION

**B.S. Computer Engineering** — GPA: 3.46 / 4.00
Khawaja Fareed University of Engineering & Information Technology (KFUEIT)

## APPLICATION QUESTIONS

**Available to start immediately:** Yes
**Comfortable working full-time, on-site in Lahore:** Yes
**Salary expectation (PKR/month):** PKR 80,000 – 120,000 (negotiable based on role scope)
**Years of experience with LLMs / Agentic AI:** 1+ year hands-on (Council AI, Line QA Bot, LLM API integrations)